

ObservedDiskElement

ObservedDiskElement [*disk*, *geodesic*, *Options*]

returns an association containing information about accretion disk '*disk*' (assoc generated by DiskParams) radiation observed on geodesic '*geodesic*' (assoc generated by KerrNullGeoDistant). The keys are 'PhysicalTemperature', 'EffectiveTemperature', 'SpecificIntensity', 'Intensity' and 'PeakFrequency'.

ObservedDiskElement[disk_Association, geodesic_, OptionsPattern[]] takes as input an association *disk*, presumably generated by the DiskParams function (but a different association with matching keys and objects as the output of DiskParams is technically possible as well), and a geodesic object, presumably generated by the KerrNullGeo or KerrNullGeoDistant functions, and an options pattern.

The following option can be given:

"Grid"	False	Specifies whether a grid of lines of constant r and φ should be generated over the data. Takes a boolean value.
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This function returns an association with the following keys:

"PhysicalTemperature"	This returns the surface temperature at the coordinates passed to the function through the geodesic object in geodesic["EmissionCoordinates"] .
"EffectiveTemperature"	This returns the temperature at which a black body with the observed intensity would radiate according to the Stefan–Boltzmann law.
"SpecificIntensity"	Returns the specific intensity calculated from disk["SpectralFluxDensity"] at the radius given by geodesic["EmissionCoordinates"] as a function of frequency, as it would be measured by the observer.
"Intensity"	Returns the integrated specific intensity given by disk["FluxDensity"]/ π at the radius given by geodesic["EmissionCoordinates"] as it would be measured by the observer.
"PeakFrequency"	Returns the peak frequency given by given by disk["PeakFrequency"] at the radius given by geodesic["EmissionCoordinates"] as it would be measured by the observer.

Tech Notes ⓘ

AlphaDiskModel

Related Links ⓘ

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See Also ⓘ

DiskParams ▪ **KerrNullGeoDistant** ▪ **KerrNullGeo** ▪ ⓘ

Related Guides

AlphaDiskModel

Examples Initialization ⓘ

Needs ["BlackHoleImages`"]

Basic Examples

[More Examples ▸](#)

Generate the functions of the accretion disk with $\alpha = 0.1$ and an influx of 10^{14} kg/s around a black hole with a solar mass and the spin parameter $a = 0.1$ using the **DiskParams** function:

```
In[32]:= disk = DiskParams[0.6, 0.1, 1500, 1]
```

Compute a geodesic in geometry given by $a = 0.6$, with the initial values $\theta = \pi/3$, $\alpha = 6$, $\beta = 7$ using the **KerrNullGeoDistant** function:

```
In[9]:= geod = KerrNullGeoDistant[0.6,  $\pi/3$ , 6, 7];
```

Generate the association of observed values:

```
In[11]:= observedElement = ObservedDiskElement[disk, geod];
```

Get the physical temperature of the observed element:

```
In[12]:= observedElement["PhysicalTemperature"]
```

```
Out[12]=  $6.6642 \times 10^6$ 
```

More Examples ⓘ

- Scope
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- "Grid"
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Metadata

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Categorization ⓘ

Keywords

Syntax Templates